12. **Dissolved Oxygen Content** Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8 (2.5)	5.2 (10)	5.0 (7.5)	4.9 (5)	5.1 (9)
Below Town	5.0 (7.5)	4.7 (1)	4.9 (5)	4.8 (2.5)	4.9 (5)

a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.

b. Use a Student's t-test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

a. H0: the two distributions are the same
H1: Oxygen levels below town are lower

$T_B = 1+2.5+5+5+7.5 = 21$ $\min(T_A, T_B) = 21$

$T_A = 2.5+10+7.5+5+9 = 34$

$n_1 = 5, n_2 = 5, \alpha = 0.05$

$T_{critical} = 19$

$21 > 19$ fail to reject H0

b. $H_0: \mu_A - \mu_B = 0$

$H_1: \mu_A - \mu_B > 0$

$\bar{x}_A = \dfrac{4.8+5.2+5+4.9+5.1}{5} = 5$

$s^2_A = 0.025$

$\bar{x}_B = \dfrac{5+4.7+4.9+4.8+4.9}{5} = 4.86$

$s^2_B = 0.0126$

$s^2_p = \dfrac{(n_a-1)s^2_A + (n_b-1)s^2_B}{n_a+n_b-2}$

$= \dfrac{4(0.025)+4(0.0126)}{8}$

$= 0.0188$

$s_p = 0.1371$

$t = \dfrac{\bar{x}_A - \bar{x}_B}{s_p\sqrt{\frac{1}{n_a}+\frac{1}{n_b}}} = \dfrac{0.14}{0.1371\sqrt{\frac{2}{5}}} = 1.6146$

$df = 5+5-2 = 8$

$t_{crit} = 1.860$

$1.6146 < 1.860$ fail to reject H0

DATA SET DS1502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

① $H0: MD - MH = 0$
$H1: MD - MH \neq 0$

② $\alpha = 0.05$

③ $TD = 126$
$TH = 45$
$T1 = 45$
$T1^* = 9(9+9+1) - 45 = 126$
$\min(T_1, T_1^*) = 45$

$n1 = 9,\ n2 = 9,\ \alpha/2 = 0.025$
$Tcrit = 62$

$T \leq Tcrit$

④ $45 \leq 62$ reject H0

15.2

Two Simple Examples *Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \le .10$. Calculate the observed value of the test statistic and present your conclusion.*

4.

	Pair						
Population	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	−	+	+	+

① $H_0: p = 0.5$
$H_1: p \neq 0.5$

② $\alpha = 0.10$

③ $n = 7$, $\#(x) = 6$

$X \sim \text{Binomial}(7, 0.5)$

$P(X \le 1) + P(X \ge 6)$

$2P(X \le 1)$

$2\left[\dfrac{\binom{7}{0} + \binom{7}{1}}{2^7}\right]$

$= 2\left(\dfrac{8}{128}\right) = \dfrac{16}{128} = 0.125$

$0.125 > 0.10$

$x = 0$ or $x = 7$

$P(x=0) + P(x=7) = \frac{1}{128} + \frac{1}{128} = 0.015625$

$0.15625 < 0.1$

$P(X \le 1) + P(X \ge 6) = 0.125$

DATA SET DS1507

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1 – 10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal		A	B	Meal		A	B
1	−	6	8	12	+	8	5
2	−	4	5	13	+	4	2
3	+	7	4	14	0	3	3
4	+	8	7	15	−	6	8
5	−	2	3	16	−	9	10
6	+	7	4	17	+	9	8
7	0	9	9	18	−	4	6
8	−	7	8	19	+	4	3
9	−	2	5	20	+	5	4
10	+	4	3	21	+	3	2
11	−	6	9	22	+	5	3

a. Use the binomial tables in Appendix I to find the exact rejection region for the test.

b. Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)

c. Compare the results of parts a and b.

$x = 11$

$n = 20$

a. $X \sim Bin(20, 0.5)$

$P(X \leq 5) = 0.021$

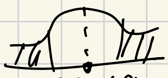
$0.021 < 0.025$

$X \leq 5$ or $X \geq 15$

11 isnt there, so fail to reject H0

b. $z = \dfrac{x - 0.5n}{0.5\sqrt{n}} = \dfrac{11 - 10}{\sqrt{5}} = \dfrac{1}{\sqrt{5}} = 0.4472$

$\alpha = 0.05$, $z = \pm 1.96$



−1.96 1.96 outside RR

fail to reject H0

C. Both methods fail to reject H0

DATA SET DS1508

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

a. Test using the sign test. Choose your rejection region so that α is near $\boxed{.05}$.

b. Why might it be inappropriate to use the Student's t-test in analyzing the data?

	Drug A			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered	
1	84	63	75.0	−
2	63	44	69.8	−
3	56	48	85.7	+
4	77	57	74.0	−
5	29	20	69.0	−
6	48	40	83.3	+
7	61	42	68.9	−
8	45	35	77.8	−
9	79	57	72.2	−
10	62	48	77.4	−

	Drug B		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

a. ① $H_0: p = 0.5$

$H_1: p \neq 0.5$

$x = 2$

$n = 10$

$X \sim Bin(10, 0.5)$

$P(X \leq 1) = 0.011$

$X \leq 1$ or $X \geq 9$

2 is outside the RR so fail to reject H0

15.4

DATA SET DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.

b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

d	Rank	signed rank
0.1	1.5	+1.5
0.7	7	+7
0.3	4	+4
−0.1	1.5	−1.5
0.5	5.5	+5.5
0.2	3	+3
0.5	5.5	+5.5

$T^+ = 1.5 + 7 + 4 + 5.5 + 3 + 5.5 = 26.5$

$T^- = 1.5$

$\min(T^+, T^-) = 1.5$

$n = 7, \ \alpha = 0.10$

$T_{crit} = 3$

$1.5 \le 3$ reject H0

b. $x = 6$

$x = 0$ or $x = 7$

since $x = 6$, fail to reject H0

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

DATA SET DS1511

Month	A	B
1	3	7
2	14	12
3	7	9
4	10	15
5	9	12
6	6	6
7	13	12
8	6	5
9	7	13

−4	4	−6
2	2	+3.5
−2	2	−3.5
−5	5	−7
−3	3	−5
0	−	
1	1	+1.5
1	1	+1.5
−6	6	−8

a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.

b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

a. $n=8$

$T^+ = 3.5+1.5+1.5 = 6.5$

$T^- = 6+3.5+7+5+8 = 29.5$

$\min(T^+, T^-) = 6.5$

$T_{crit} \approx 4$

$T \leq 4$

$6.5 > 4$

fail to reject H_0

b. if monthly conditions differ
- diff production workloads or operating hours
- Maintenance schedules
- diff operators or raw material quality
- Temperature and environmental conditions

DATA SET DS1514

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5	11	17	8
2	24	9	12	20	16
3	20	5	13	20	10
4	18	9	14	16	12
5	22	6	15	24	7
6	19	11	16	22	9
7	20	8	17	25	21
8	19	11	18	21	14
9	17	7	19	19	12
10	21	9	20	23	13

a. What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?

b. Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.

c. Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

a. 1. paired t-test
2. sign test
3. Wilcoxon signed rank test

For paired t-test, the paired differences should:
- be independent from student to student
- be approximately normally distributed
- have no serious outliers

then, we calculate
d = with − without
the differences range from 4 to 17 appear reasonably symmetric & have no obvious extreme outliers.
Therefore, the paired t-test assumptions appear reasonable

b. sign test
① $H_0: p = 0.5$
$H_1: p \neq 0.5$
② $n = 20, x = 20$
$x \leq 5$ or $x \geq 15$

$20 \geq 15$
reject H0

Wilcoxon signed rank test
1. H_0: the 2 recall score distributions are the same
H_1: the 2 recall score distributions are different

$T^+ = \frac{20(21)}{2} = 210$

$T^- = 0$

$\min(T^+, T^-) = 0$

$T^- \leq T_{crit}$
$0 \leq 52$
reject H0

c. Both reject the H0, so they have same conclusion that imagery instructions significantly improve recall scores

15.5

Applying the Basics

DATA SET DS1518 **6. Legos®** The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)			
.5	1.0	1.5	2.0
8 13	9 16	4 2	4 2
14 20	7 10	6 7.5	7 10
9 16	5 5	7 10	5 5
12 19	8 13	8 13	5 5
10 18	9 16	6 7.5	4 2

Use the Kruskal–Wallis H-Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

	Training Period (hours)				
	.5	1.0	1.5	2.0	
	8 13	9 16	4 2	4 2	
	14 20	7 10	6 7.5	7 10	
	9 16	5 5	7 10	5 5	
	12 19	8 13	8 13	5 5	
	10 18	9 16	6 7.5	4 2	
T_i	86	60	40	24	210

$k=4$
$n=5$

H0: there is no diff in the distribution of times for the four diff lengths
H1: at least one of the distribution is different

$\alpha = 0.01$

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{T_i^2}{n_i} - 3(n+1)$$

$$= \frac{12}{20(21)} \left(\frac{86^2}{5} + \frac{60^2}{5} + \frac{40^2}{5} + \frac{24^2}{5} \right) - 3(21)$$

$H = 12.269$

$df = k-1 = 4-1 = 3$
for $\alpha = 0.01$ & $df = 3$
$\chi^2 crit = 11.345$

$12.269 > 11.345$
reject H0

DATA SET DS1520

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10–19	20–39	40–59	60–69
	29 21	24 8	37 39	28 18
	33 29.5	27 15	25 10.5	29 21
	26 12.5	33 29.5	22 5.5	34 34
	27 15	31 24	33 29.5	36 37.5
	39 40	21 3	28 18	21 3
	35 36	28 18	26 12.5	20 1
	33 29.5	24 8	30 23	25 10.5
	29 21	34 34	34 34	24 8
	36 37.5	21 3	27 15	33 29.5
	22 5.5	32 25.5	33 29.5	32 25.5
Total	309 247.5	275 168	295 216.5	282 188

$k=4$
$n=10$

820

a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal–Wallis H-test with $\alpha = .01$.

b. Find the approximate p-value for the test in part a.

c. Since the F-test in Chapter 11 and the H-test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F-test produced the test statistic $F = 0.87$ with p-value $= 0.468$, compare this p-value with the p-value in part b and explain the implications of the comparison.

H0: the four age group heart rate have the same location
H1: at least two of it differend in location

a. $H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$

$= \frac{12}{40(41)} \left(\frac{247.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right) - 3(41)$

$H = 2.632$

$df = k-1 = 4-1 = 3$

$\alpha = 0.01$

χ^2 critical $= 11.345$

$2.632 < 11.345$

fail to reject H0

b. $H = 2.647$, $df = 3$

$P \approx 0.449$

$0.449 > 0.01$

fail to reject H0

c. $F = 0.87$, P value $= 0.468$

$0.468 > 0.01$

fail to reject H0

Both tests fail to reject H0

10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.[7] The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

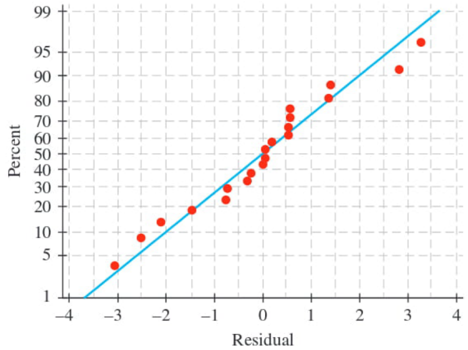
DATA SET DS1528

Block Child	Antibiotic Treatment 1	2	3	4
1	4.8 2	2.2 1	6.8 4	6.2 3
2	8.1 2	9.2 3	6.6 1	9.6 4
3	5.0 3	2.6 1	3.6 2	6.5 4
4	7.9 2	9.4 4	5.3 1	8.5 3
5	3.9 3	7.4 4	2.1 2	2.0 1
	12	13	10	15

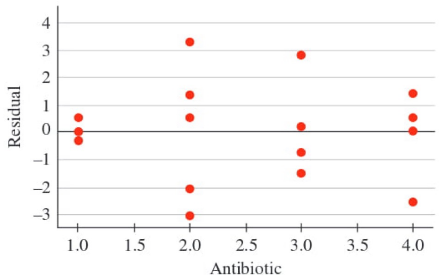
a. What design is used in collecting these data?

b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?

Normal Probability Plot
(response is Score)



Residuals Versus Antibiotic
(response is Score)



c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

a. randomized block design
treatments: the four antibiotics
Blocks: the five children
each child rates all four antibiotics
Bcs the same children test every antibiotic, the observations are related, not independent. the friedman test is the non parametric equivalent of a randomized block design

b. the normal probability plot is approximately linear, so the normality assumption appears reasonable
However, the residual plot shows different amounts of variability for the four antibiotics. In particular, antibiotic 1 has much less variation than the other
Therefore, the equal variance assumption doesn't appear satisfied

c. $\alpha = 0.05$
H0: the response distributions are the same for all four antibiotics
H1: at least one antibiotic has a different response distribution

$$F_r = \frac{12}{bk(k+1)} \sum T_i^2 - 3b(k+1)$$

$$= \frac{12}{20(5)} \sum (12^2+13^2+10^2+15^2) - 3(5)(5)$$

$F_r = 1.56$

$df = k-1 = 4-1 = 3$

$\chi^2_{0.05,3} = 7.815$

$1.56 < 7.815$
fail to reject H0

$p \approx 0.668$
$0.668 > 0.05$
fail to reject H0

11. Yield of Wheat Exercise 9 (Chapter 11 Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

DATA SET DS1529

Treatment Variety	Location BLOCK 1	2	3	4	5	6	
A	35.3 4	31.0 2	32.7 3	36.8 3	37.2 3	33.1 3	18
B	30.7 1	32.2 3	31.4 2	31.7 2	35.0 2	32.7 2	12
C	38.2 5	33.4 4	33.6 4	37.1 4	37.3 4	38.2 5	26
D	34.9 3	36.1 5	35.2 5	38.3 5	40.2 5	36.0 4	27
E	32.4 2	28.9 1	29.2 1	30.7 1	33.9 1	32.1 1	7

a. Use the Friedman F_r-test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.

b. When the analysis of variance was performed in Chapter 11, the ANOVA F-test produced the test statistic $F = 18.61$ with p-value $= .000$. How do these results compare with results of the Friedman F_r-test in part a? Explain.

a. Friedman Fr-test

1. Ho: The yield for the five different varieties of wheat is the same
   H1: At least one wheat variety has different yield
2. $\alpha = 0.05$
3. $Fr\ STAT = \frac{12}{BK(K+1)} \sum Ti^2 - 3b(k+1) \sim \chi^2(k-1)$

$Fr\ stat = \frac{12}{30(6)} (18^2 + 12^2 + 26^2 + 27^2 + 7^2) - 3(6)(6)$

$Fr^* = 20.133$

$Fr\ crit = \chi^2_{0.05, 4} = 9.488$

$20.133 > 9.488$

reject H0

b. $F = 18.61$
$Fr = 20.133$
both bigger than 9.488
so same, reject Ho

19.7

DATA SET DS1530

11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating		Distance		d	d^2
1	12	7.5	75	3	4.5	20.25
2	7	4	165	7	−3	9
3	5	3	300	12	−9	81
4	19	12	15	1	11	121
5	17	10	180	8	2	4
6	12	7.5	240	11	−3.5	12.25
7	9	6	120	4	2	4
8	18	11	60	2	9	81
9	3	1	230	10	−9	81
10	8	5	200	9	−4	16
11	15	9	130	5.5	3.5	12.25
12	4	2	130	5.5	−3.5	12.25

$\Sigma d^2 = 454$

a. Calculate Spearman's rank correlation coefficient r_s.

b. Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

a. $r_s = 1 - \dfrac{6 \Sigma d_i^2}{n(n^2-1)}$

$r_s = 1 - \dfrac{6(454)}{12(12^2-1)}$

$r_s = -0.587$

b. $H_0: \rho_s = 0$

$H_1: \rho_s < 0$

$\alpha = 0.05$

r_s crit $= -0.503$

rejection region $= r_s \leq -0.503$

$r_s \approx -0.587 < -0.503$

reject H_0

$p \approx 0.021 < 0.05$ reject H_0

DATA SET DS1531

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:[8]

Runner	Years of Competitive Running		10-Kilometer Finish Time (minutes)		d	d^2
1	9	7	33.15	1	6	36
2	13	9	33.33	2	7	49
3	5	3.5	33.50	3	0.5	0.25
4	7	6	33.55	4	2	4
5	12	8	33.73	5	3	9
6	6	5	33.86	6	-1	1
7	4	2	33.90	7	-5	25
8	5	3.5	34.15	8	-4.5	20.25
9	3	1	34.90	9	-8	64

$\Sigma d^2 = 208.5$

a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.

b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

a. $r_s = 1 - \dfrac{6\Sigma d_i^2}{n(n^2-1)}$

$= 1 - \dfrac{6(208.5)}{9(80)}$

$r_s = -0.738$

b. $H_0: \rho_s = 0$

$H_1: \rho_s \neq 0$

$\alpha = 0.05$

$r_s = -0.738$

$n = 9$, $\alpha = 0.05$, two tailed

r_s crit $= 0.683$

$|R_s| \geq 0.683$

-0.683 0.683

$0.738 \geq 0.683$

reject H_0